

Commentary

Waste heat is mostly a waste of time

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Introduction

Heat accounts for three-quarters of industrial emissions, and decarbonizing industrial heat is essential not only to reduce emissions but to secure the long-term competitiveness of manufacturing economies. Addressing this challenge demands a shift toward scalable, electrified, and high-efficiency heat supply technologies that can replace fossil combustion at process-relevant temperatures.¹

Efforts to harness waste heat in industrial settings have spanned decades, animated by the promise of squeezing more value out of fuel already burned. In principle, waste heat recovery improves process energy efficiency and reduces emissions.² In practice, however, the returns are rarely worth the investment, especially at low temperatures. Since at least the mid-20th century,³ engineers have pursued two main strategies: converting waste heat into power or using it to upgrade heat via industrial heat pumps.

Despite steady research interest, widespread commercial success in using waste heat to generate power remains elusive. A recent technoeconomic analysis (TEA) by Geffroy et al. underscored why: when waste heat is below 100°C, the capital costs and low conversion efficiencies of power generation technologies (e.g., thermoelectrics, organic Rankine cycles)

make them economically unviable.⁴ The numbers simply don't add up.

This commentary extends that argument to waste heat upgrading via heat pumps. Using a simple TEA framework, we show that waste heat upgrading via industrial heat pumps is also rarely viable at low temperatures. The additional capital expense required to extract and raise the temperature of such heat is significant—and often more than can be justified by the modest efficiency gains. The problem is not one of physics but one of engineering economics and deployment logistics. Waste heat, especially at low temperatures, tends to be diffuse, intermittent, and inconveniently located. This turns every integration effort into a bespoke project, limiting scalability and making standardization—arguably the most powerful force in modern energy innovation—all but impossible.

Background

The key insight of Geffroy et al. was that waste heat conversion to electricity is only viable when the waste heat source is both hot and consistent. For sources below 100°C, the Carnot efficiency of conversion is low, and the capital cost per unit of electricity generated is prohibitively high.⁴ Their analysis considered multiple technology pathways and

concluded that none were able to achieve reasonable payback periods under practical conditions.

What about using waste heat to supply or supplement thermal energy needs, rather than converting it to electricity? Specifically, what about industrial heat pumps that upgrade waste heat to process-relevant temperatures, like saturated steam? On paper, this seems more efficient: the second law of thermodynamics is less punitive when heat is moved rather than converted, but the TEA tells a familiar story.

Waste heat upgrading systems require a heat engine in reverse: a heat pump. This introduces capital costs associated with compressors, power electronics, and especially heat exchangers. Critically, capturing low-grade heat requires very large surface areas to overcome low temperature differentials, driving up heat exchanger size and cost.

Even worse, every facility is different. Waste heat comes in different forms (e.g., flue gas, jacket cooling water), at different temperatures, from different locations, and with different duty cycles. Integration therefore becomes a custom-engineering problem. Off-the-shelf solutions don't exist. The result is one-off system designs, high engineering costs, and costly interruptions to



Table 1. Qualitative comparison of electric solutions for decarbonizing industrial heat

Characteristic	Air-source heat pump	Waste heat heat pump	Electric boiler
Installation complexity	low to moderate	high	low
Footprint	moderate (air coils)	moderate to large (waste heat capture)	small
COP/operating expenses (OPEX)	moderate (1.7 to >2)	moderate to high (1.7 to >4)	low (~1)
CAPEX	low	high	low
Potential for operational disruption	low	high	low

manufacturing operations. For a **technology to scale, it must be replicable**. Waste heat is the antithesis of replicable.

An **alternative** to waste heat is to upgrade ambient air to the required process temperature in an **air-source heat pump** (Table 1). Similar to an electric boiler, this approach is promising because it can be a **drop-in installation for heat decarbonization projects** that does not rely upon complex integration requirements associated with waste heat-driven heat pumps. The coefficient of performance (COP) of an air-sourced system is generally **lower** than systems that utilize waste heat, which is why so many are focused on using waste heat. However, as we show below in a simple example, **waste heat is not free**, and it is often unattractive relative to drop-in solutions due to unfavorable project economics.

A simple technoeconomic illustration

A simple TEA illustrates the burden that waste heat integration poses on industrial heat decarbonization projects. Here we compare multiple approaches to electrifying the **150°C steam boiler** of a manufacturing facility with \$100 million per year in revenue and a **1 megawatt (MW) thermal load**.

In this analysis, we compare installing an **electric resistive boiler** (today's *de facto* electrification solution) with (1) a **productized and standardized air-sourced steam heat pump** and (2) a **waste heat-sourced steam heat pump**. The air-sourced steam heat pump, as its name implies, sources and upgrades heat from the ambient air outside of the manufacturing facility, while the waste heat-sourced steam heat pump takes advantage of existing waste heat in the facility and upgrades it to generate steam,

thus increasing operational efficiency of the system. For the waste heat-sourced system, different source temperatures are considered, including both **COP dependence on source temperature and capital expenditure (CAPEX) requirements for different temperature systems** (e.g., for lower-temperature waste heat, a multi-stage heat pump system will be required, while for higher temperature waste heat, a single-stage heat pump is required, thus decreasing CAPEX).⁵

In this analysis, we use data from the Energy Information Administration (EIA) on **CAPEX costs for waste heat integration projects including heat recovery equipment**, associated labor and materials, project and construction costs, and overall project engineering and fees as shown in Table 2 adapted from Darrow et al.⁶ and EIA.⁷

In Figure 1A, a side-by-side comparison of the **associated project costs (stacked bars)** and **annual operating costs (line)** are plotted for each of the systems considered for 150°C steam-generated with electricity at a standard US industrial electricity rate of **\$80/MWh**. One notable difference between **waste heat-driven systems** and the electric and air-sourced systems is the need to account for facility **downtime** necessary to install **additional heat exchangers and fluid lines** in the manufacturing floor to capture and collect waste heat. In this analysis, both **0 and 5 days of downtime** are considered, and the associated lost revenue for 5 days of downtime is shown here.

As illustrated by Figure 1A, **heat pump systems represent much lower operational cost** compared to resistive technology but much **higher CAPEX** as well. The simple **payback period** is used by industry to estimate how quickly a capital project is

“paid for” by operational savings and can be calculated as the **difference in capital cost between two options divided by the difference in annual operating cost**.

$$\text{Payback Period}[\text{yr}] = \frac{\Delta \text{CapEx}[\$]}{\Delta \text{OpEx} \left[\frac{\$}{\text{yr}} \right]}$$

For example, in this scenario, the **payback period of the air-sourced heat pump relative to the electric boiler** can be calculated to be $(\$1,133,000 - \$260,000) / (\$715,919 - \$404,044) = 2.8 \text{ years}$. **Waste heat-driven heat pumps** can also have **similar payback periods** relative to electric resistance boilers, but these **increase substantially when facility downtime is considered**. For the representative \$100 million/year revenue facility, **5 days of downtime nominally doubles the payback period** in each case. For example, although a heat pump that upgrades 60°C waste heat to 150°C has a **higher COP (2.73)** than an air-sourced system (1.73), the **payback without downtime is still higher than for an air-source system (3.6 vs. 2.8 years)**, and it increases to 6.5 years if the facility stops operating for a mere 5 days.

To further illustrate the benefit of lower overall integration costs, Figure 1B shows the incremental payback period for each of the waste heat systems relative to an air-sourced heat pump system. For each temperature, the bars represent the **range for the payback period from 0 days of downtime (low) to 5 days of downtime (high)**.

Clearly, low-temperature waste heat <30°C offers little benefit to an air-sourced system, with payback periods >10 years. There are **efficiency gains with higher temperature waste heat**, but any **facility downtime** does not justify capturing it when an **air-source heat pump is available with lower integration costs**. Looking at 60°C, the payback period can approach **15 years** relative to an air-sourced system if the facility is down for only **5 days**. Moreover, this waste heat might not even be **available** when steam is needed for the process, pushing the payback even further out by potentially adding **thermal storage or, for example, redundant electric resistance boilers**. These results underscore a painful truth: industrial waste heat recovery

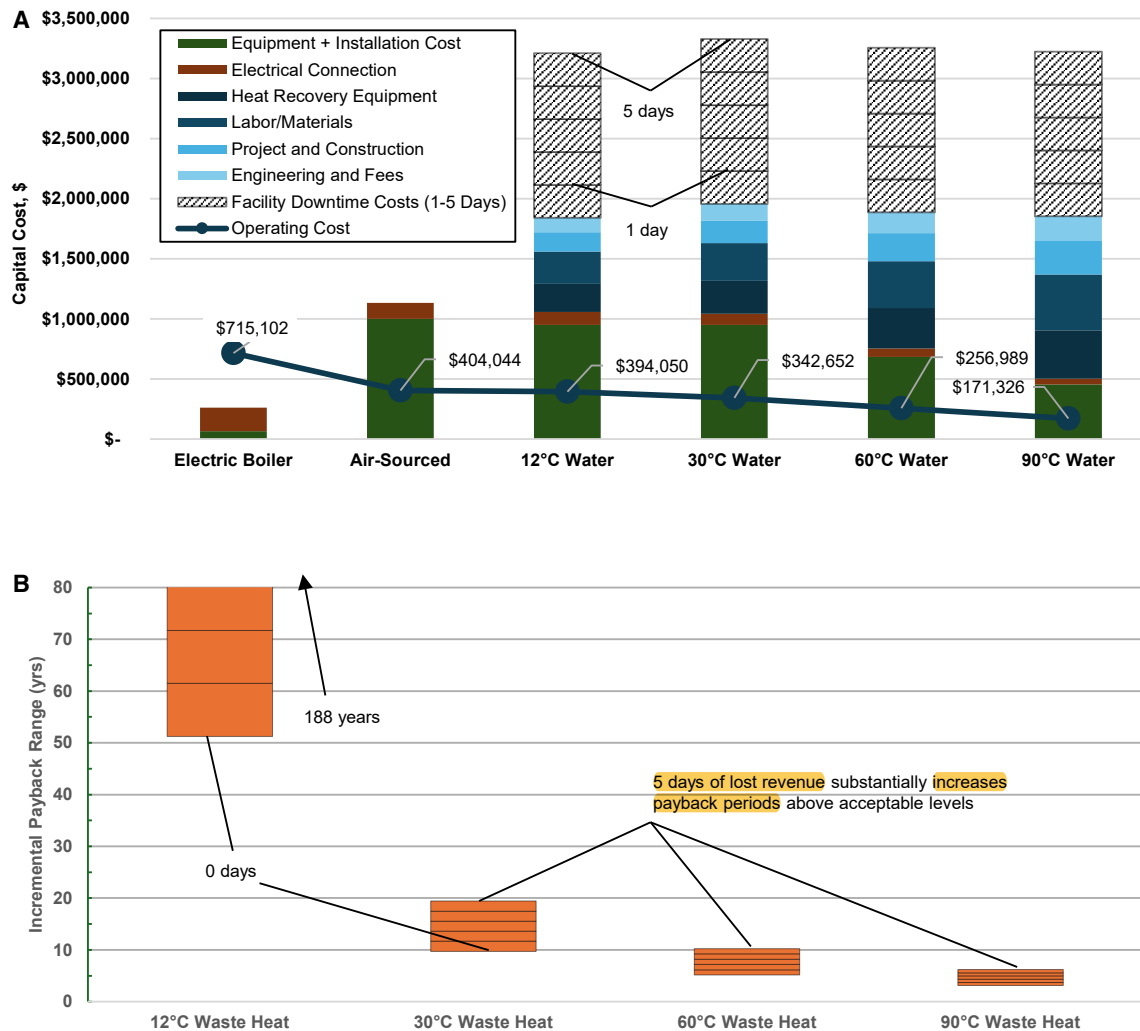


Figure 1. Economic comparison for a 150°C steam boiler electrification project in a \$100 million/year revenue facility with \$80/MWh electricity

(A) Raw comparison of incremental CAPEX costs (stacked bars) include lost revenue for associated with 1- 5-day downtime to install waste heat capture systems on manufacturing floor. OPEX costs (line) are the associated annual electricity cost to operate the system.

(B) Incremental payback period range of waste heat-driven heat pump systems compared to the air-sourced system to deliver 150°C steam including 0 to 5 days installation downtime.

projects are **expensive, slow to pay back, and highly situational**. Even when source temperatures are favorable, the lack of modularity and need for bespoke engineering undercut economic viability.

Why waste heat won't scale

There is a fundamental difference between technologies that **scale through economies of scale** and those that **scale through economies of mass manufacturing**. Boilers, for example, have been **standardized and mass produced** for over 160 years. The steam boiler is one of the oldest examples of replicable industrial

infrastructure. This standardization has enabled **rapid adoption, low costs, and widespread support infrastructure**.

The same story has played out with solar photovoltaics and lithium-ion batteries: technologies that are mass manufactured, modular, and easily deployed. These systems don't require custom engineering for every installation. They scale not because each individual system is huge but because **manufacturing and deployment are streamlined**.

Waste heat integration, by contrast, **scales** only through **customized projects**—akin to building a new chemical

plant every time a system is installed. If we want to **electrify heat at scale**, we must prioritize technologies that are **modular, replicable, and factory built**. Air-source heat pumps fit this mold. Waste heat systems do not.

Beyond technoeconomics: Practical considerations

There are **additional hurdles** to waste heat utilization beyond pure economics. Examples are as follows:

- **Spatial** constraints: installing **large heat exchangers** and new fluid

Table 2. Associated project cost assumptions for waste heat integration

Category	Cost	Source description
Heat recovery	\$532/kW	the heat recovery equipment consists of the exhaust economizer that extracts heat from the exhaust system, process heat exchanger for extracting heat from the engine jacket coolant, circulation pump, control system, and piping
Labor/materials	\$614/kW	labor costs for civil, mechanical, and electrical work; material costs for ductwork, piping, wiring
Project and construction	\$369/kW	general contractor markup and bonding, performance guarantees
Engineering and fees	\$274/kW	project engineering and fees
Total heat recovery costs (T > 30 C)	\$1,788/kWth	–
Total heat recovery costs (T < 60 C.)	\$1,341/kWth	25% reduction based on team experience
Interconnect/electrical	\$192/kWe	required electrical interconnect costs for electric boilers and heat pumps

circuits on a crowded factory floor is not trivial.

- **Temporal** availability: industrial processes require on-demand heat at specified flow rates. Waste may not be available at the same time process steam is required.
- **Regulatory** burdens: custom installations may trigger new permitting requirements or environmental impact assessments.
- **Maintenance** complexity: a non-standard system is harder to service and often lacks trained technicians and spare parts.
- **Risk** aversion: manufacturers are conservative. A retrofit that introduces downtime or unproven technologies faces a high barrier to adoption.

This is not to say that waste heat has no role. In some specific high-temperature, high-duty-cycle applications, especially when heat is already cleanly separated, conveniently located, and available when

steam is needed, there may be economic justification.² However, these are exceptions, not the rule. The industrial sector is massive and diverse. We must pursue solutions that work broadly.

Conclusion: Focus on solutions that scale

Industrial heat accounts for a significant portion of global emissions, and decarbonizing it is essential for reaching net-zero goals, but we must be honest about what technologies can scale.

The appeal of waste heat recovery—getting something for nothing—is seductive but misleading. In practice, waste heat is often diffuse, messy, and expensive to harness. The resulting systems are complex, bespoke, and slow to deploy.

If we want to decarbonize industrial heat quickly and cost effectively, we should direct our resources and attention toward scalable, replicable, and modular solutions. That means prioritizing stan-

dardized factory-built technologies, it means electrifying heat directly where possible, and it means saying “no” to elegant engineering that fails the real-world test of deployment at scale.

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DECLARATION OF INTERESTS

A.K.S., A.S., and T.B. are co-founders and shareholders of AtmosZero, Inc., an air-sourced steam heat pump start-up focused on developing drop-in steam heat pump technology that does not require waste heat.

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